

Tab 1



National Teachers College
629 Nepomuceno St., Quiapo, Metro Manila
School of Arts, Sciences, and Technology (SoAST)

Final Project: Open Access Resources Tracker (OART)

2.7 BSIT

Submitted By:

Alonzo, Christian P.

Cabug, Jerome E.

Saad, Moh Ben Amroden P.

Macahia, Ralph Steven

Geraldo, Adrianne James

Submitted To :

Prof. Ms. Justin Louise R. Neypes



1. Introduction

Project Overview

The group developed the Open Access Resources Tracker (**OART**), a VB.NET Windows Forms application based on United Nations Sustainable Development Goal 4: Quality Education. The system is designed to help students organize, manage, and access learning materials in a more efficient and organized way.

The application functions as a complete desktop-based system with a login module, CRUD operations, database integration using ADO.NET and SQL Server, and reporting features. The system aims to provide students with easier access to educational resources while promoting organization and productivity.

UN SDG Target

The Open Access Resources Tracker supports United Nations Sustainable Development Goal 4: Quality Education. SDG 4 aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.



By providing students with an organized platform to manage and access learning materials, OART directly contributes to improving access to educational resources. The system promotes self-directed learning and makes it easier for students to keep track of their study materials, supporting the broader goal of quality education for everyone.

Connection of the System to the UN SDG

The Open Access Resources Tracker is directly connected to SDG 4: Quality Education because it addresses the challenge of organizing and accessing learning materials efficiently. Students often struggle to manage multiple resources across different subjects, which can hinder their learning.

OART solves this by providing a centralized platform where students can add, view, edit, and delete learning resources. Features like search, filtering, and report generation make it easier for students to find what they need and track their progress. This demonstrates how technology can support quality education in a practical and accessible way.

2. Requirement Analysis

This section presents the functional and non-functional requirements of the Open Access Resources Tracker. These requirements define the system's expected behavior, features, and



quality attributes necessary to ensure effective management of learning resources aligned with SDG 4: Quality Education.

A. Functional Requirements

Functional requirements specify the essential system functions that allow users and administrators to interact with the Open Access Resources Tracker.

FR-1 — User Registration and Authentication

The system shall allow users to register and log in securely. Only authenticated users shall be permitted to manage and access learning resources within the system.

FR-2 — Resource Management (CRUD)

The system shall enable users to add, view, update, and delete learning resources. Each resource shall include a title, subject, type, description, URL or file path, and status.

FR-3 — Resource Categories

The system shall support categorization of resources by subject area and type (e.g., PDF, Video, Link, Article) to help users organize their learning materials effectively.

FR-4 — Search and Filtering



The system shall provide search and filtering functions that allow users to quickly find resources by title, subject, type, or status.

FR-5 — Input Validation and Error Handling

The system shall validate all user inputs and display appropriate error messages to guide users in providing correct and complete information.

FR-6 — Report Generation

The system shall generate organized reports of learning resources using a built-in reporting tool, allowing users to view and export resource summaries.

FR-7 — User Role Management

The system shall support user roles to control access levels. Administrative users shall have additional privileges such as managing all records in the system.

B. Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints under which the system must operate.

NFR-1 — Usability

The system shall provide a clean, user-friendly interface that students can navigate easily with minimal technical knowledge.



NFR-2 — Performance

The system shall process requests and retrieve data from the database within an acceptable response time under normal operating conditions.

NFR-3 — Reliability

The system shall ensure data accuracy and integrity. All resource records and user information shall be stored reliably without data loss.

NFR-4 — Security

The system shall implement secure login mechanisms to protect user accounts and restrict unauthorized access to the system's features and data.

NFR-5 — Maintainability

The system shall be designed with organized, well-commented source code to allow easy maintenance, debugging, and future enhancements.

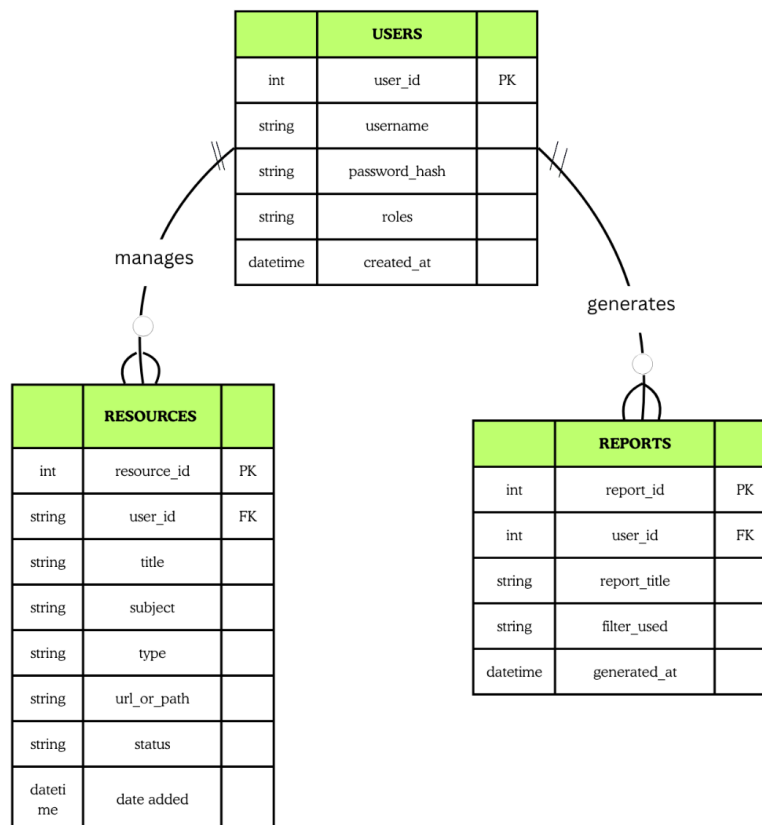
NFR-6 — Compatibility

The system shall run on Windows-based machines using VB.NET and SQL Server, ensuring compatibility with standard academic computing environments.



3. Database Design

Entity Relationship Diagram (ERD)





Data Dictionary (Table Names, Fields, Types, and Constraints)

Table: USERS

Field	Type	Key	Constraints	Description
user_id	INT	PK	NOT NULL, AUTO_INCREMENT	Unique identifier for each user
username	VARCHAR(50)		NOT NULL, UNIQUE	Login username
password_hash	VARCHAR(255)		NOT NULL	Hashed password string
role	VARCHAR(20)		NOT NULL, DEFAULT 'user'	Access role: admin or user
created_at	DATETIME		NOT NULL, DEFAULT NOW()	Account creation timestamp

Table: RESOURCES

Field	Type	Key	Constraints	Description
resource_id	INT	PK	NOT NULL, AUTO_INCREMENT	Unique identifier for each resource
user_id	INT	FK	NOT NULL, REFERENCES USERS	Owner/uploader of the resource
title	VARCHAR(150)		NOT NULL	Name/title of the learning material
subject	VARCHAR(100)		NOT NULL	Subject area (e.g. Math, Science)
type	VARCHAR(50)		NOT NULL	Resource type: PDF, Video, Link, etc.
url_or_path	VARCHAR(500)		NULLABLE	URL or file path to the resource
description	TEXT		NULLABLE	Short description of the resource
status	VARCHAR(20)		DEFAULT 'active'	Status: active or archived
date_added	DATETIME		NOT NULL, DEFAULT NOW()	Timestamp when resource was added

Table: REPORTS

Field	Type	Key	Constraints	Description
report_id	INT	PK	NOT NULL, AUTO_INCREMENT	Unique identifier for each report
user_id	INT	FK	NOT NULL, REFERENCES USERS	User who generated the report
report_title	VARCHAR(150)		NOT NULL	Title or name of the report
filter_used	VARCHAR(200)		NULLABLE	Filters/criteria applied (e.g. by subject)
generated_at	DATETIME		NOT NULL, DEFAULT NOW()	When the report was generated



4. System Architecture

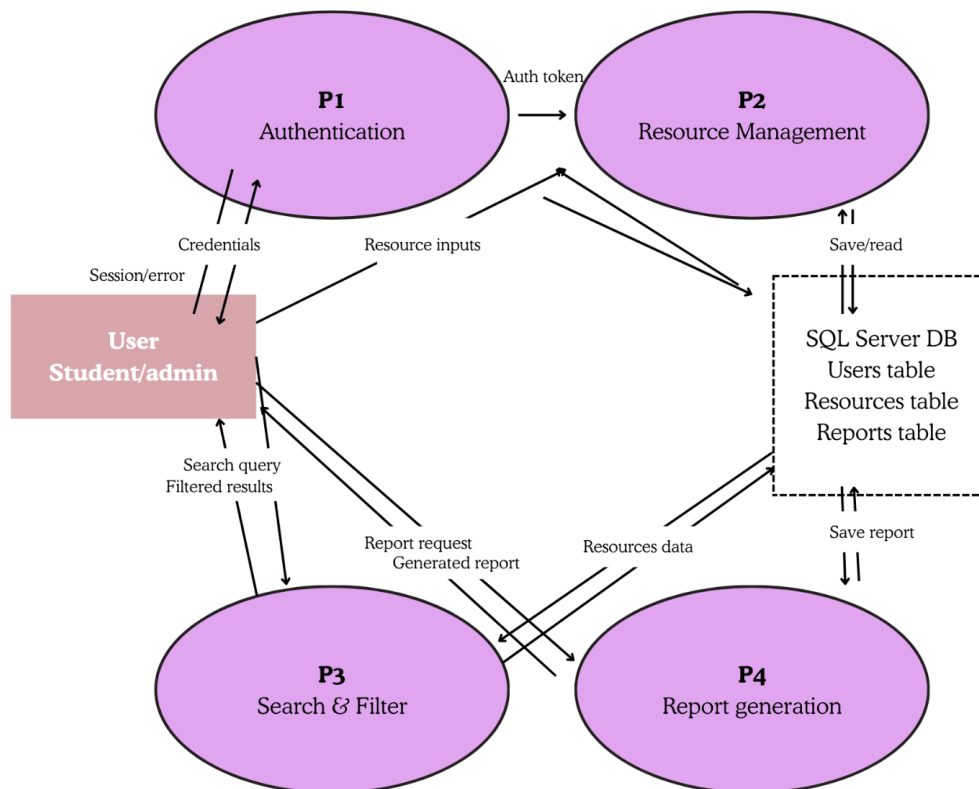
Data Flow

The system architecture of OART follows a straightforward data flow where users interact with the Windows Forms interface, which communicates with the database through ADO.NET. All resource data, user information, and report records are stored in and retrieved from SQL Server.

DFD Level 0 and Level 1



DFD Level 0 - Context Diagram



DFD Level 1 - Internal system processes



5. Individual Contribution

The following table details the specific modules and forms implemented by each group member throughout the development of the Open Access Resources Tracker.

Member	Role	Modules / Forms Implemented
Alonzo, Christian P.	Project Lead	Login form, session management, database connection class (ADO.NET), overall system architecture, and GitHub repository setup.
Cabug, Jerome E.	Backend Dev	CRUD operations — AddResourceForm, EditResourceForm, delete logic, input validation, and error handling.
Saad, Moh Ben Amroden P.	UI/UX Dev	MainForm dashboard layout, SearchFilterForm, ViewResourceForm, and overall navigation and interface refinement.
Macahia, Ralph Steven B.	Reports & Docs	ReportViewerForm, ERD, flowcharts, data dictionary, SDAD documentation, system testing, and final submission preparation.
Geraldo, Adrienne James	QA & Testing	System testing and



National Teachers College
629 Nepomuceno St., Quiapo, Metro Manila
School of Arts, Sciences, and Technology (SoAST)

debugging, performance
evaluation, identifying and
documenting bugs, and
verifying that all features
meet the project
requirements before final
submission.

Tab 2

